Preemptive and non preemptive in os in simple and easy way

In Operating Systems (OS), preemptive and non-preemptive refer to how the CPU is given to processes (programs).

Think of the CPU like a teacher helping students one by one.

1️⃣ Preemptive Scheduling

Definition:

The OS can take the CPU away from a running process and give it to another process if needed.

Simple Idea:

A process can be interrupted before it finishes.

Example:

You are typing in Word, and suddenly a notification appears or you open Chrome.

The OS stops Word for a moment and runs Chrome.

Example in real life:

A teacher is helping student A, but if student B has an urgent question, the teacher pauses A and helps B first.

Common Algorithms

- Round Robin
- Shortest Remaining Time First
- Priority Scheduling (preemptive)

Advantages

- Faster response time
- Better for multitasking systems

2️⃣ Non-Preemptive Scheduling

Definition:

Once a process starts using the CPU, it cannot be stopped until it finishes or voluntarily gives up the CPU.

Simple Idea:

A process runs completely before another starts.

Example:

If one program starts running, all other programs must wait until it finishes.

Real life example:

A teacher is helping student A and will not stop until A is fully done, even if others are waiting.

Common Algorithms

- First Come First Serve (FCFS)
- Shortest Job First (non-preemptive)
- Priority Scheduling (non-preemptive)

Advantages

- Simple to implement
- Less overhead

🔑 Quick Difference

Feature	Preemptive	Non-Preemptive
CPU interruption	Allowed	Not allowed
Multitasking	Better	Poor
Response time	Fast	Slow
Example	Round Robin	FCFS

✅ One-line memory trick:

- Preemptive = CPU can be taken away anytime

- Non-Preemptive = CPU stays with process until it finishes

If you want, I can also show a very easy exam example with a timeline diagram (teachers love asking that).

Like an interrupt signal

Yes, very similar to an interrupt signal 👍

Preemptive Scheduling

In preemptive OS, the CPU can stop a running process using something like an interrupt signal.

- The OS sends a timer interrupt or another interrupt.
- The current process is paused.
- Another process is given the CPU.

Example

Process A is running → interrupt occurs → OS stops A → Process B runs.

So you can think:

✅ Preemptive = CPU can interrupt a running process

Non-Preemptive Scheduling

In non-preemptive OS, the running process will not be interrupted by the OS.

- The process keeps running
- It stops only when it finishes or voluntarily gives up CPU

Example

Process A starts → it runs until completion → then Process B starts.

So:

❌ No interrupt to take CPU away

Easy Way to Remember

Type	Interrupt Used?	Behavior
Preemptive	✅ Yes	Process can be stopped
Non-Preemptive	❌ No	Process runs until finished

✅ One simple sentence for exams:

Preemptive scheduling allows the OS to interrupt a running process and give the CPU to another process, while non-preemptive scheduling does not allow interruption until the process finishes.